

DEMAND-AWARE PRICING SIMULATION FOR THE NYC YELLOW TAXI SYSTEM

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Abstract

Urban transportation systems operate under dynamic conditions shaped by demand, traffic congestion, airport activity, and weather. However, regulated taxi systems such as the New York City Yellow Taxi network rely on mostly static fare structures. This project develops a demand-aware pricing simulation framework that combines NYC Yellow Taxi trip data with weather information to predict fares and simulate bounded pricing adjustments. Three machine learning models—Ridge Regression, Random Forest, and XGBoost—were evaluated, with XGBoost producing the best results. The final system applies interpretable adjustments for demand, traffic, weather, and airport conditions while keeping changes transparent. The framework is implemented through an interactive Streamlit prototype with fare estimation, route preview, and a natural-language pricing assistant, showing how machine learning can support fair, interpretable, and regulator-friendly taxi pricing decisions.

1 INTRODUCTION

Urban mobility systems are highly dynamic. In New York City, taxi demand changes by hour, location, weather, airport activity, and traffic conditions. However, the NYC Yellow Taxi system operates under a regulated fare structure that does not fully adapt to these real-world conditions. This creates a mismatch between operating difficulty and fare pricing.

For example, a taxi trip during heavy rain at 5 PM may involve higher demand, slower traffic, and longer trip duration, but the fare structure remains largely fixed. In contrast, a similar trip during normal conditions may require less driver effort and less travel time. Static pricing supports predictability, but it does not fully reflect the operational differences between these scenarios.

Ride-sharing platforms such as Uber and Lyft address supply-demand imbalance through surge pricing. However, unrestricted surge pricing can create fairness concerns, especially during emergencies, bad weather, or high-demand events. A regulated taxi system cannot simply copy ride-sharing surge pricing because fairness, transparency, and public trust are essential.

This project explores a middle-ground approach: bounded, transparent, and explainable pricing adjustments. The goal is not to create aggressive surge pricing. Instead, the project builds a data-driven framework that predicts fares and applies small, interpretable adjustments based on demand, traffic, weather, and airport-related factors.

The project also converts the analytical framework into an interactive system. The final prototype allows users to estimate fares, view route context, and ask natural-language questions about pricing. This makes the work not only a modeling exercise, but also a decision-support system for regulated transportation pricing.

2 PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

The NYC Yellow Taxi fare structure does not dynamically respond to changing demand, traffic, and weather conditions. Static pricing supports fairness and predictability, but it may also reduce efficiency during high-demand or high-cost operating periods.

This creates two connected problems. First, fares may not reflect the true operating difficulty of certain trips, especially during rush hour, congestion, airport travel, or bad weather. Second, riders

may not understand why fares are high or low unless the system provides a clear explanation. Therefore, the challenge is to design a pricing framework that improves efficiency while remaining bounded, transparent, and understandable.

2.2 Objectives

The main objectives of this project are:

1. Predict taxi fares using historical NYC Yellow Taxi trip records.
2. Engineer demand, traffic, temporal, and weather-related features.
3. Compare Ridge Regression, Random Forest, and XGBoost models.
4. Build a bounded pricing simulation layer.
5. Evaluate revenue impact compared to static pricing.
6. Develop an interactive Streamlit prototype with fare breakdown, route preview, and chatbot explanation.

3 DATA SOURCES

3.1 NYC Yellow Taxi Dataset

The primary dataset is the NYC Yellow Taxi trip records for January 2024. The dataset contains approximately 2.9 million taxi trips. Important fields include pickup and dropoff time, trip distance, pickup and dropoff location IDs, passenger count, fare amount, total amount, surcharges, payment type, tolls, tips, congestion surcharge, and airport fee.

This dataset is useful because it captures real trip behavior across different times, locations, and fare conditions. It also provides the spatial information needed to estimate demand patterns at the pickup-zone level.

3.2 Weather Dataset

The weather dataset contains daily weather observations for New York City. Features include temperature, precipitation, snow, snow depth, humidity, and wind gust. Weather data was merged with taxi trips using pickup date.

Weather was included because it affects both rider behavior and operating conditions. Rain or snow can increase taxi demand because fewer riders may choose to walk, bike, or use exposed transit options. At the same time, bad weather can reduce vehicle speed and increase trip duration.

3.3 Dataset Integration

The taxi data was trip-level and timestamp-based, while the weather data was daily. The integration was performed using the pickup date. This allowed each taxi trip to receive weather context for the day on which it occurred.

Although hourly weather would provide more detail, daily weather still provides useful information for identifying bad-weather days and evaluating their effect on fare behavior and pricing adjustments.

4 DATA PREPROCESSING

4.1 Large-Scale Handling

Since the full taxi dataset contains millions of records, chunk-based loading was used to avoid memory issues. The data was processed in chunks, and a representative sample of approximately 200,000 trips was retained for modeling.

This approach allowed the project to work within practical computing limits while still preserving enough data volume for training and evaluation. The sampled dataset included trips across different pickup hours, zones, trip distances, and fare values.

4.2 Cleaning Rules

The dataset contained invalid or extreme values. The following cleaning filters were applied:

- Trip distance greater than 0.01 miles and less than or equal to 60 miles.
- Trip duration greater than 1 minute and less than or equal to 180 minutes.
- Total fare amount greater than \$0.01 and less than or equal to \$300.
- Average speed between 0.5 mph and 80 mph.

These filters removed unrealistic trips and improved model reliability. For example, zero-distance trips, negative fares, impossible speeds, and extremely long durations can distort model training. Cleaning therefore helped ensure that the model learned realistic taxi fare patterns rather than data-entry errors.

4.3 Target Variable

The target variable for fare prediction was total amount. This value reflects the final amount associated with a trip, including fare-related components and surcharges. Predicting total amount is useful because it represents what the system ultimately needs to estimate for a user-facing fare prediction tool.

5 FEATURE ENGINEERING

Feature engineering was one of the most important parts of the project because raw trip records do not directly describe demand pressure, congestion, or weather severity.

5.1 Temporal Features

Temporal features were derived from the pickup timestamp, including pickup hour, pickup weekday, weekend indicator, rush-hour indicator, and night indicator. These features allow the model to capture daily travel patterns such as commuter peaks, nightlife demand, and lower overnight demand.

To capture the cyclical nature of time, pickup hour was also encoded using sine and cosine transformations:

$$hour_sin = \sin\left(\frac{2\pi h}{24}\right) \quad (1)$$

$$hour_cos = \cos\left(\frac{2\pi h}{24}\right) \quad (2)$$

This representation helps capture the fact that late-night hours such as 23:00 and 00:00 are close in real time, even though their raw numeric values are far apart.

5.2 Demand Index

Demand was calculated at the pickup-zone and hour level. This was done because demand in Midtown Manhattan at 5 PM is very different from demand in an outer borough at the same time.

$$DemandIndex_{z,h} = \frac{Trips_{z,h}}{MedianTrips_z} \quad (3)$$

A value above 1 indicates higher-than-usual demand for that pickup zone and hour. This feature is important because it localizes demand instead of treating New York City as one uniform market.

5.3 Traffic Proxy

Since live traffic data was not available, trip speed was used as a proxy for congestion. Traffic intensity was derived from trip speed relative to a reference speed of 15 mph and clipped between 0 and 1.

$$TrafficIntensity = clip\left(\frac{15 - speed}{15}, 0, 1\right) \quad (4)$$

Lower speeds indicate heavier traffic and higher traffic intensity. This feature helps capture the operational burden of congested trips, where a driver may spend more time completing the same distance.

5.4 Weather Features

A bad-weather flag was created using precipitation, wind gust, snow, and snow depth. The training pipeline marked a day as bad weather when precipitation, wind gust, snow, or snow depth crossed predefined thresholds.

Weather variables were not used only as raw numeric inputs. They were also converted into a practical indicator because the pricing layer needs simple and interpretable conditions. This makes the system easier to explain to riders and regulators.

5.5 Interaction Features

Interaction features were also created, including demand multiplied by traffic and demand multiplied by weather severity. These interactions help capture combined effects, such as high demand during bad weather or high demand during congestion.

For example, rain alone may increase demand slightly, and rush hour alone may increase demand slightly. However, rain during rush hour can produce a stronger combined effect. Interaction features help the model represent this type of combined pressure.

6 METHODOLOGY

6.1 Model Selection

Three models were compared:

- **Ridge Regression:** a regularized linear baseline.
- **Random Forest:** a nonlinear tree-based ensemble.
- **XGBoost:** a gradient-boosted tree model suited for structured tabular data.

These models were selected because they represent increasing levels of complexity and allow comparison between simple linear modeling and stronger nonlinear methods.

Ridge Regression provides a useful baseline because it is easy to interpret and helps measure whether the engineered features have linear predictive power. Random Forest improves on this by capturing nonlinear relationships through multiple decision trees. XGBoost was included because gradient boosting often performs strongly on structured tabular datasets and can capture complex feature interactions.

6.2 Training and Testing

The cleaned dataset was split into training and testing sets. The models were trained using engineered trip, temporal, demand, traffic, weather, and location features. Model performance was evaluated on held-out test data.

The use of a held-out test set helped ensure that model performance reflected generalization rather than memorization. Since taxi trips vary significantly by location and time, evaluating on unseen examples was important for estimating real-world usefulness.

6.3 Evaluation Metrics

The main evaluation metrics were:

- **RMSE:** penalizes large prediction errors more strongly.
- **MAE:** measures average absolute error.
- R^2 : measures how much variance the model explains.

RMSE was treated as the primary metric because large fare errors are more harmful in a pricing system. A few large errors can reduce trust more than many small errors. MAE was also useful because it provides an easier-to-understand average dollar error.

7 MODEL PERFORMANCE

Table 1. Model Performance on Test Data

Model	RMSE	MAE	R^2
Ridge Regression	4.62	2.18	0.91
Random Forest	3.27	1.34	0.96
XGBoost	2.89	1.01	0.981

XGBoost achieved the best performance with the lowest RMSE and MAE and the highest R^2 . This result is expected because taxi fare prediction depends on nonlinear relationships among distance, duration, demand, traffic, location, weather, and time.

The RMSE of 2.89 indicates that the model generally predicts fares within a small dollar range. This is strong performance for a real-world transportation dataset, where fare variation can be affected by many factors such as route choice, congestion, tolls, airport fees, and local surcharges.

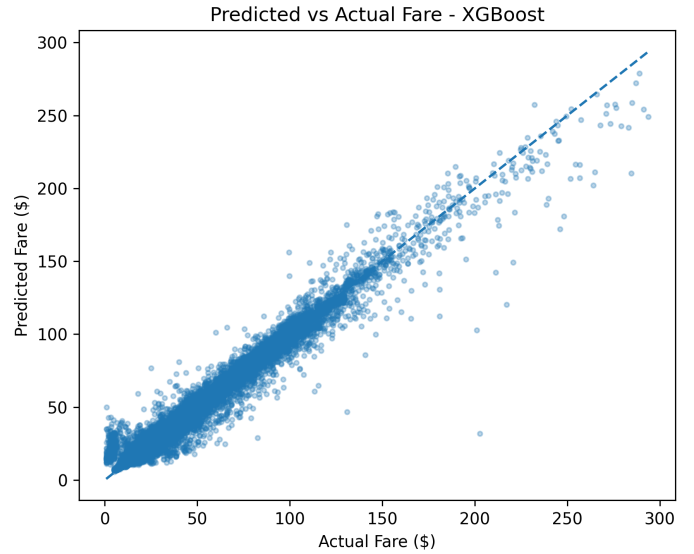


Fig. 1. Predicted versus actual fares for the XGBoost model.

8 DEMAND MODELING

Demand was modeled using zone-hour aggregation rather than a separate deployed time-series forecasting model. For each pickup zone and hour, trip counts were calculated and compared against the median hourly trip count for that zone. This produced a DemandIndex value that indicates whether demand was higher or lower than usual for that specific zone and hour.

This approach allowed the pricing simulation to respond to localized demand pressure instead of treating the entire city as one demand region. This is especially important in New York City because demand is highly spatial. A high-demand condition in Midtown does not necessarily mean demand is high in every borough.

The demand heatmap shows how trip activity changes across hours and high-volume pickup zones. Peak patterns are important because they support the logic behind demand-aware pricing adjustments.

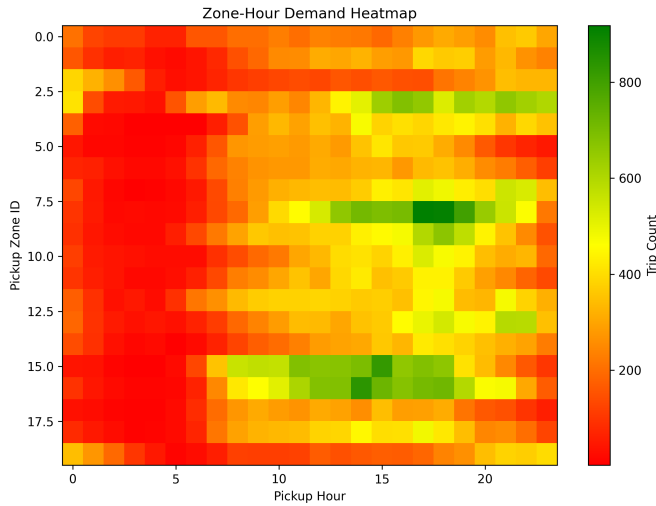


Fig. 2. Zone-hour demand heatmap showing taxi demand variation across pickup hours and high-volume pickup zones. Warmer colors indicate higher demand intensity.

9 PRICING SIMULATION FRAMEWORK

9.1 Purpose

The pricing simulation was designed to test whether small and explainable adjustments can improve revenue without using aggressive surge pricing. The system starts with the predicted base fare and then applies bounded, rule-based adjustments.

The key idea is to separate machine learning prediction from policy-based adjustment. The machine learning model estimates the base fare, while the simulation layer applies transparent adjustment rules. This makes the pricing framework easier to explain and audit.

9.2 Adjustment Factors

The implemented pricing engine uses four adjustment factors:

- **Demand adjustment:** applied when DemandIndex is at least 1.4.
- **Traffic adjustment:** applied during high traffic intensity.
- **Weather adjustment:** applied during bad-weather conditions.
- **Airport adjustment:** applied for airport-related trips.

Table 2. Implemented Pricing Adjustment Components

Condition	Adjustment
DemandIndex $\geq$ 1.4	\$1.00
High traffic intensity	\$1.25
Bad-weather day	\$0.50
Airport trip	\$0.75
Post-adjustment multiplier	1.03

This pricing layer is not an official TLC fare rule. It is a simulation layer created for evaluating how demand, traffic, weather, and airport conditions could influence fares in a controlled and interpretable way.

9.3 Why Bounded Adjustments Matter

Bounded adjustments are important because they prevent the system from behaving like unrestricted surge pricing. Each adjustment is small, explicit, and tied to a condition. This supports transparency and makes the framework more suitable for regulated transportation.

From a rider perspective, the fare increase is easier to accept if the reason is clearly shown. From a regulator perspective, the adjustment logic can be audited because it is rule-based rather than hidden inside a black-box decision.

10 RESULTS AND ANALYSIS

10.1 Revenue Impact

The combined pricing simulation produced an estimated revenue lift of approximately 2.6% compared to the baseline static-pricing estimate. This result shows that even small, bounded adjustments can produce measurable revenue improvement while remaining more transparent and controlled than unrestricted surge pricing.

A 2.6% improvement may appear modest, but it is meaningful at taxi-system scale because millions of trips occur over time. More importantly, the improvement is achieved without extreme multipliers or unpredictable fare spikes.

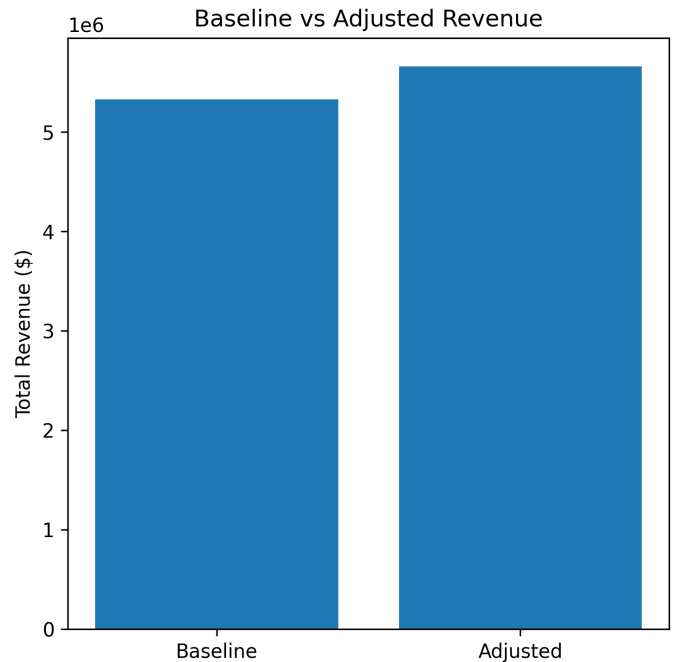


Fig. 3. Comparison of baseline and adjusted revenue showing approximately 2.6% increase.

10.2 Traffic Analysis

Traffic affects taxi economics because congested trips take longer and reduce the number of trips a driver can complete in the same period. Since live traffic data was not available, the system used trip speed to estimate traffic intensity.

The traffic intensity plot shows the relationship between speed and congestion score. Lower speeds correspond to higher traffic intensity, while speeds above the reference threshold receive lower congestion pressure.

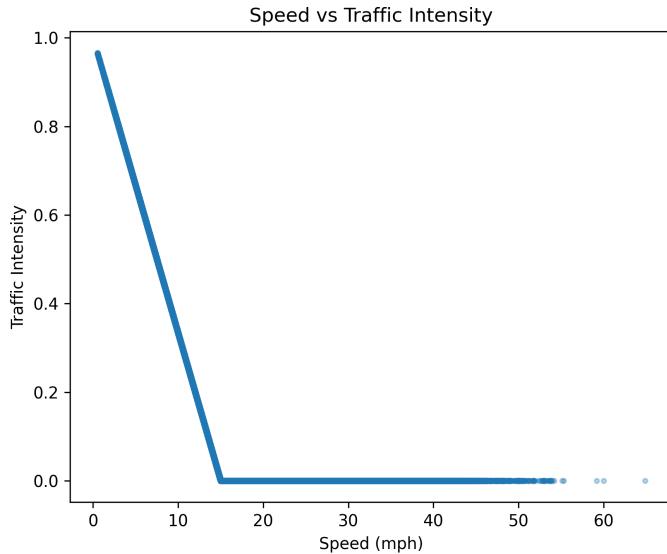


Fig. 4. Relationship between trip speed and traffic intensity. Lower trip speeds correspond to higher traffic intensity.

11 SYSTEM IMPLEMENTATION

11.1 Streamlit Application

The final prototype was built using Streamlit. It provides two main tabs: a fare estimator and a pricing assistant. The fare estimator allows users to select pickup and dropoff TLC zones, choose pickup hour and weekday, set passenger count, select weather condition, and estimate the fare. The system displays a base model prediction, final estimated fare, estimated duration, price change, and a detailed breakdown of demand, traffic, weather, and airport adjustments.

The application converts the analytical model into a usable interface. This is important because a pricing framework is only useful if users can interact with it and understand its outputs.

11.2 Fare Estimator Interface

The fare estimator connects the trained XGBoost model with the rule-based adjustment layer. In the example shown, the system estimates a trip from Times Square/Theatre District to Green-Wood Cemetery during bad weather. The output includes the estimated fare, base fare, duration, price change, and individual adjustment values.

The interface separates the base model prediction from the adjusted fare. This distinction makes it clear which part comes from machine learning and which part comes from the pricing policy simulation.

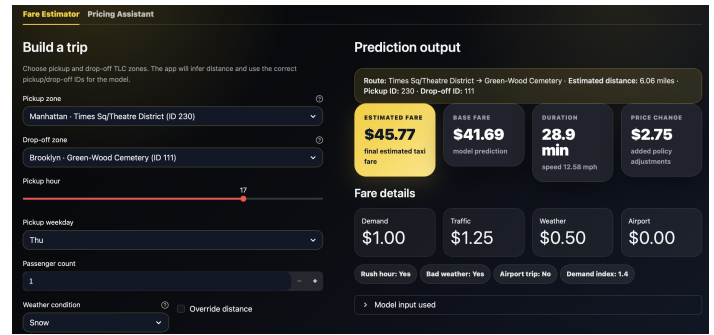


Fig. 5. Fare estimator interface showing zone selection, weather condition, predicted fare, base fare, duration, and pricing adjustment breakdown.

11.3 Route Preview

The application also includes a route preview map. When a pickup and dropoff zone are selected, the prototype visualizes the route on a dark map interface and reports estimated road distance and duration. This helps users understand the spatial context behind the fare prediction.

Route visualization improves the interpretability of the fare estimate because it connects the numeric output to a physical trip path. This is especially useful when users compare trips between different boroughs or airport zones.

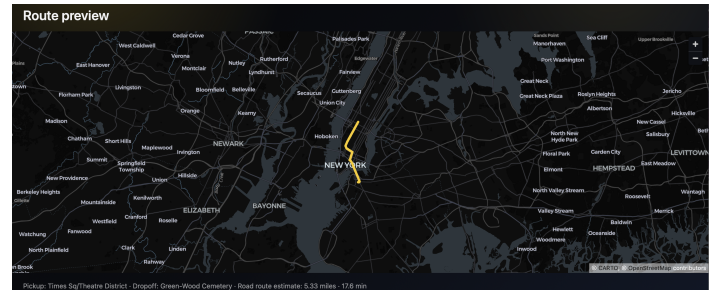


Fig. 6. Route preview showing the estimated trip path, pickup zone, dropoff zone, estimated distance, and duration.

11.4 Pricing Assistant

The pricing assistant supports natural-language project and trip questions. It can answer service-availability questions, explain pricing behavior, and suggest cheaper travel times. For example, the assistant identifies that service is available to Newark Airport and recommends off-peak travel times to reduce the estimated fare for a Manhattan-to-JFK trip.

This assistant improves accessibility because users do not need to understand the technical feature engineering or model pipeline. Instead, they can ask practical questions and receive plain-language explanations.

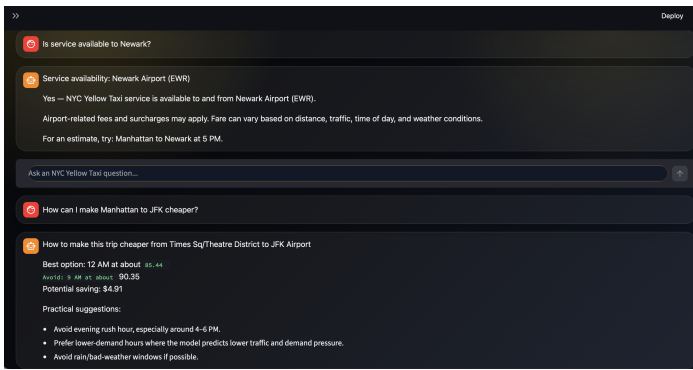


Fig. 7. Pricing assistant showing natural-language responses for service availability and cheaper travel-time recommendations.

12 EXPLAINABILITY

The system provides rule-based explanations for pricing adjustments. It does not use formal explainable-AI methods such as SHAP or LIME. Instead, each adjustment is explicitly tied to a clear condition, such as rush-hour demand, bad weather, traffic intensity, or airport travel. This makes the output easier for users and regulators to understand.

Explainability is a major part of this project because dynamic pricing systems often fail when users do not understand why prices changed. In this framework, every adjustment is shown directly in the interface. This makes the system more transparent than black-box surge pricing.

13 SYSTEM-LEVEL IMPACT

13.1 Passenger Perspective

From a passenger standpoint, the system maintains fairness by ensuring that pricing adjustments are bounded and transparent. Unlike unrestricted surge pricing, the adjustment amounts are small and clearly connected to real-world conditions.

The natural-language assistant also helps passengers understand pricing behavior. For example, a passenger can ask why a fare is higher at 5 PM or whether traveling later would reduce cost.

13.2 Driver Perspective

Driver-level earnings data was not available, so direct driver income impact could not be measured. However, the proposed adjustments may indirectly support drivers by increasing compensation during difficult operating conditions such as traffic congestion, bad weather, or high demand.

This is important because congested trips reduce the number of rides a driver can complete per hour. A small traffic-related adjustment helps reflect this operational burden.

13.3 Regulatory Perspective

The framework is designed for a regulated taxi environment. Unlike surge pricing, it does not apply large multipliers. Instead, it uses predefined and auditable adjustment rules.

This makes the system more suitable for regulators because the logic is transparent and can be modified before deployment. A regulator could adjust thresholds, caps, or conditions and simulate outcomes before applying any policy.

13.4 Economic Efficiency

The approximate 2.6% revenue lift demonstrates that small adjustments can still improve system efficiency. The result suggests that a regulated taxi system does not need aggressive pricing changes to achieve measurable benefits.

The project therefore supports a balanced approach: improve revenue and operational alignment while preserving fairness and public trust.

14 KEY FINDINGS

The key findings from this project are:

- XGBoost performed best among all tested models.
- Distance and duration were the strongest fare-related predictors.
- Demand was modeled at the pickup-zone and hour level.
- Traffic was represented using a speed-based proxy.
- Weather data was integrated as daily contextual information.
- The pricing simulation produced approximately 2.6% revenue lift.
- The Streamlit system converted the model into an interactive decision-support prototype.
- The pricing assistant improved transparency by explaining fare changes and suggesting lower-cost travel options.

15 LIMITATIONS

This project has several limitations. First, driver-level earnings data was not available, so direct driver income impact could not be measured. Second, weather data was daily rather than hourly, which limits storm-level precision. Third, traffic was estimated using trip speed instead of live traffic data. Fourth, route distances in the prototype are estimates based on zone and route information rather than official TLC meter readings. Fifth, the framework was evaluated through simulation rather than real-world deployment. Finally, because the taxi data focused on January 2024, the results may not fully generalize to all seasons.

Another limitation is that rider behavior is not modeled. In a real deployment, passengers may change travel time, choose another mode, or cancel trips if prices change. This behavioral response is outside the scope of this project but would be important for policy deployment.

16 FUTURE WORK

Future work could improve the system by adding live traffic APIs, hourly weather data, and driver-level earnings data. A separate demand forecasting model could also be developed using zone-hour historical demand patterns and lagged demand features. This would allow the system to predict upcoming demand rather than relying only on observed zone-hour demand indices.

The prototype could also be expanded into a regulator-facing dashboard where different pricing policies could be simulated before real-world deployment. Additional improvements could include more precise route-distance APIs, stronger chatbot grounding, and fairness analysis across boroughs and pickup zones.

A fairness analysis would be especially important because pricing changes should not disproportionately affect specific neighborhoods or passenger groups. Future versions could compare adjustment distributions by borough, zone, time, and trip type.

17 CONCLUSION

This project demonstrates that static taxi pricing can be improved using machine learning and bounded pricing adjustments. By combining fare prediction, demand modeling, traffic estimation, weather integration, route preview, and natural-language explanation, the proposed framework provides a fair and transparent alternative to unrestricted surge pricing.

The main contribution is not only the fare prediction model, but the complete decision-support system that connects prediction, pricing simulation, route-level interaction, and explanation for a regulated urban transportation environment.

The results show that even conservative pricing adjustments can improve revenue while remaining interpretable. This supports the broader idea that transportation systems can become more adaptive without abandoning regulatory fairness.

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